Práctica cero: Mecánica pulmonar

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Table of Contents

Información general....1
Datos de la simulación....1
Respuestas: Hipotenso, Normotenso e Hipertenso....2
Respuesta: Hipotenso, Normotenso e Hipertenso....2
Respuesta:Hipotenso, Normotenso e Hipertenso....3
Modelo de EDOs....4
Funcion: Respuestas de la simulacion 4

Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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Datos de la simulación

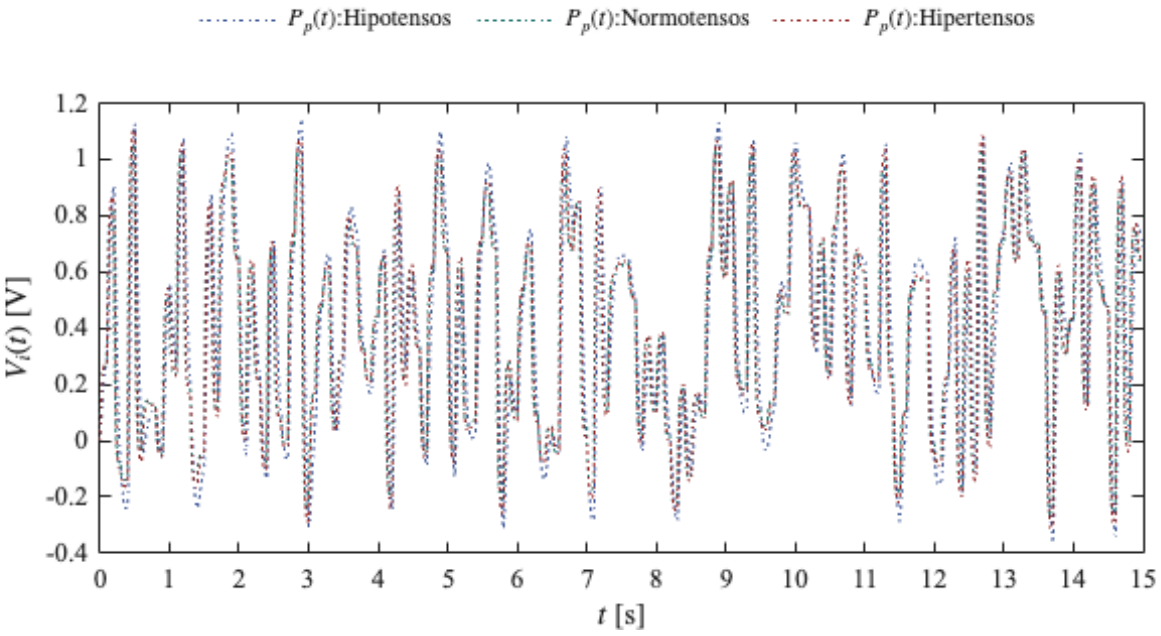
```
clc; clear; close all; warning('off','all')

open_system('hipertenso.slx')

parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
x = sim('hipertenso', parameters);
t_data=(0:0.1:15)';
t=(0:0.001:15)';
Pa = interp1(t_data,x.Pa,t,'previous');
writematrix(x.Pa,'signal.xlsx')
```

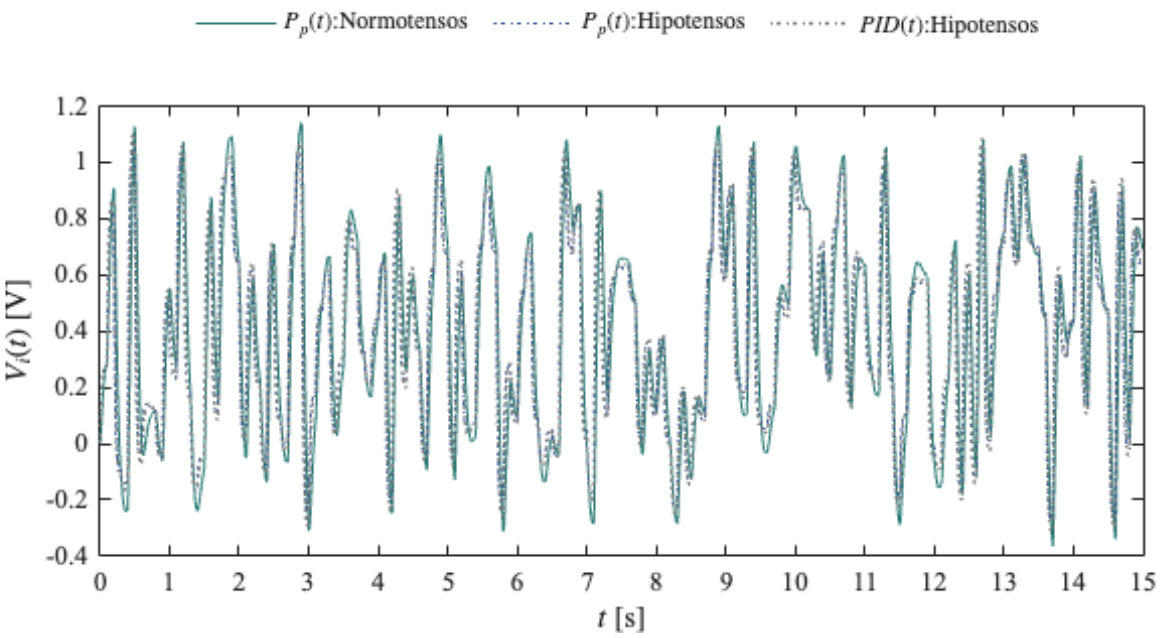
Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL, x.PpN, x.PpH, 'individuos')
```



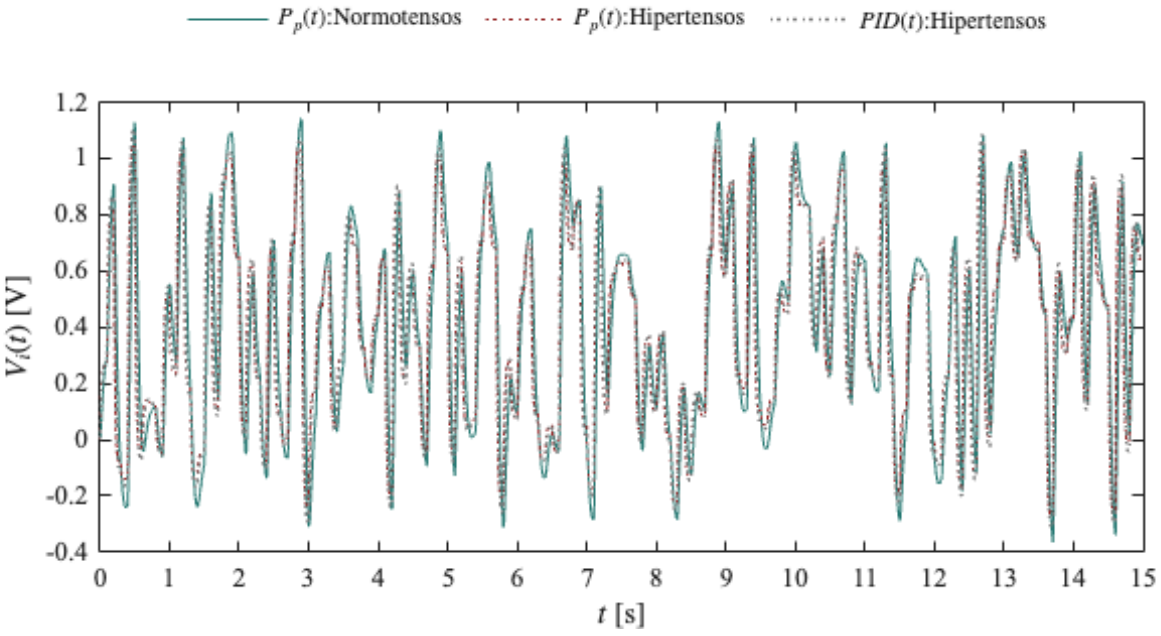
Respuesta: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL, x.PpN, x.PpH, 'hipotenso')
```



Respuesta:Hipotenso, Normotenso e Hipertenso

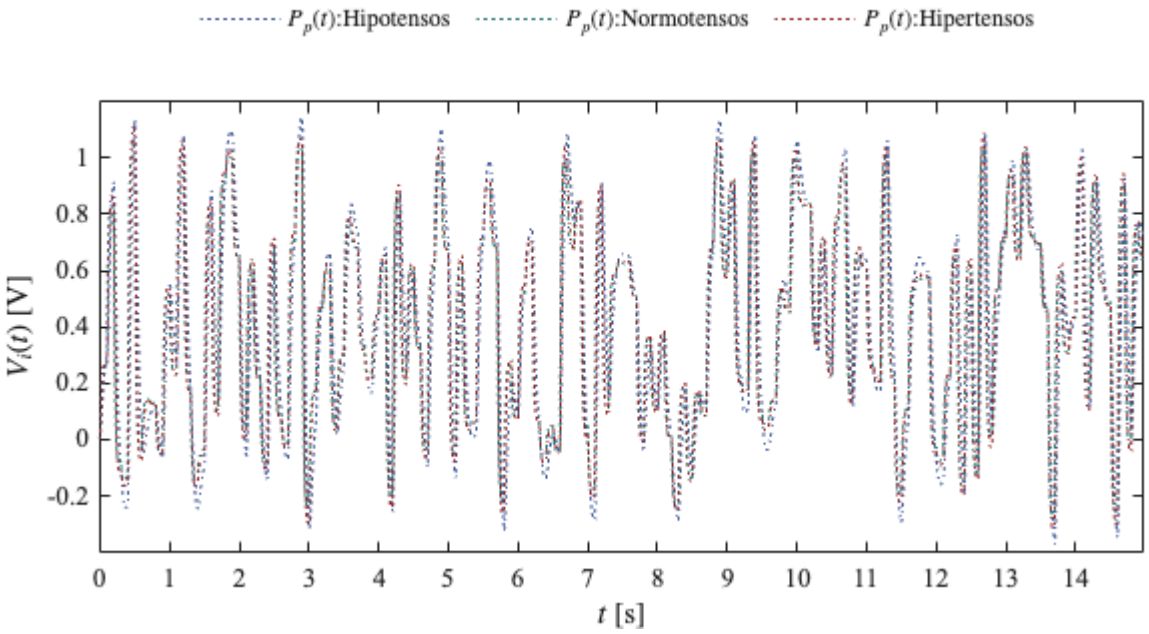
```
plotsignals(x.t,x.PpL, x.PpN, x.PpH, 'hipertenso')
```



Modelo de EDOs

```
par_hipo=[0.01,0.001,1.00,3.500]; %[Z,L,R,C]
par_normo=[0.01,0.001,1.00,1.500];
par_hiper=[0.01,0.001,1.00,1.000];

PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'odes')
```




```
function Pp= cardio(par,Pa,t)
   Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
   n= round (t(end)/ dt);
   FL= zeros(length(t),1);FL(1)=0;
   Pp= zeros(length(t),1);Pp(1)=0;
   for i= 1:n
       FL(i+1) = FL(i) + (-Pp(i)/L + Pa(i)/L)*dt;
       Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/
(Z*C))*dt;
 end
end
```

Funcion: Respuestas de la simulacion

```
function plotsignals(t,x1,x2,x3,caso)
set(figure(),'color','w')
set(gcf,'units','inches','position',[1,1,7,3.5])
set(gca,'fontname','times new roman','fontsize',10)
hold on; grid off; box on;
clr1 = [66, 91, 154]/255;
clr2 = [40, 122, 116]/255;
clr3 = [151, 40, 40]/255;
clr4 = [0.5, 0.5, 0.5];

if caso == "individuos"
    plot(t,x1,':','LineWidth',0.25,'Color',clr1)% Hipotenso
    plot(t,x2,':','LineWidth',0.25,'Color',clr2)% Normotenso
    plot(t,x3,':','LineWidth',0.25,'Color',clr3)% Hipertenso
    L = legend('$P_p(t)$:Hipotensos','$P_p(t)$:Normotensos','$P_p(t)
$:Hipertensos');
elseif caso == "hipotenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr2)% Normotenso
    plot(t,x2,':','LineWidth',0.25,'Color',clr1)% Hipotenso
    plot(t,x3,':','LineWidth',1,'Color',clr4)% Controlador del hipotenso
    L = legend('$P_p(t)$:Normotensos','$P_p(t)$:Hipotensos','$PID(t)
$:Hipotensos');
elseif caso == "hipertenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr2)% Normotenso
    plot(t,x2,':','LineWidth',0.25,'Color',clr3)% Hipertenso
    plot(t,x3,':','LineWidth',1,'Color',clr4)% Controlador del hipertenso
    L = legend('$P_p(t)$:Normotensos','$P_p(t)$:Hipertensos','$PID(t)
$:Hipertensos');
elseif caso == "odes"
        plot(t,x1,':','LineWidth',0.25,'Color',clr1)% Hipotenso
        plot(t,x2,':','LineWidth',0.25,'Color',clr2)% Normotenso
        plot(t,x3,':','LineWidth',0.25,'Color',clr3)% Hipertenso
        L = legend('$P_p(t)$:Hipotensos','$P_p(t)$:Normotensos','$P_p(t)
$:Hipertensos');
end

set(L,'interpreter','latex','location','northoutside','orientation','horizont
al','box','off')
xlabel('$t$ [s]','interpreter','latex')
ylabel('$V_i(t)$ [V]','interpreter','latex')

xlim([0 15]); xticks(0:1:15)
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
end
```